

HW5 Butterworth Filter Design Flow

1.

Here we intend to design a discrete-time low-pass filter according to a continuous-time Butterworth filter. Once the cut-off frequency is set as 0.2π , a helpful transformation named bilinear transformation maps this frequency to discrete-time domain. The relationship of frequency between in discrete-time domain and in continuous-time domain can be described as

$$\Omega = \frac{2}{T_d} \tan(\omega/2)$$

where Ω and ω represent frequency in discrete-time domain and continuous-time domain respectively, and T_d is a sampling interval which samples the impulse response (or filter here).

Once again, the goal is to devise a discrete-time low-pass Butterworth filter with cut-off frequency at 0.2π . Wherever the cut-off frequency is in s domain or z domain it should be 0.2π as much as possible, so that

$$T_d = \frac{2}{\Omega_c} \tan(\Omega_c/2)$$

For the sake of stability, all poles are at the left half of the s-plane and uniformly distributed in angle on a circle of radius Ω_c . Then bilinear transformation is implemented to map these poles to the left half of z-plane. A powerful function *poly* is used to build two polynomials referring to these poles or zeros, i.e. numerator and denominator of the filter. Finally, The resultant function, FunctionForQ1, returns DEN and NUM for any given N and cut-off frequency.

```
function [DEN,NUM] = FunctionForQ1(N,Omega_c)
% N : filter order
% Omega_c : cut-off frequency in continuous-time domain

Td = 2*tan(Omega_c/2)/Omega_c;
% uniformly distributed angles
angle = pi/2 + (0.5 : 1 : N-0.5).*pi./N;
s_p = Omega_c*exp(1j*angle); % look up end of text book
% map s to z
z_p = (1+Td*s_p/2)./(1-Td*s_p/2); % Bilinear Transform
DEN = poly(z_p); % Denominator A(z)

z_zero = -1*ones(1,N);
NUM = poly(z_zero); % Numerator B(z)

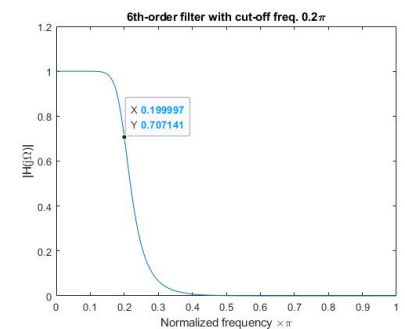
DCgain = 2^N/sum(DEN); % B(z)/A(z) evaluated at z = 1, i.e. @ omega = 0.
NUM = NUM/ DCgain;

end
```

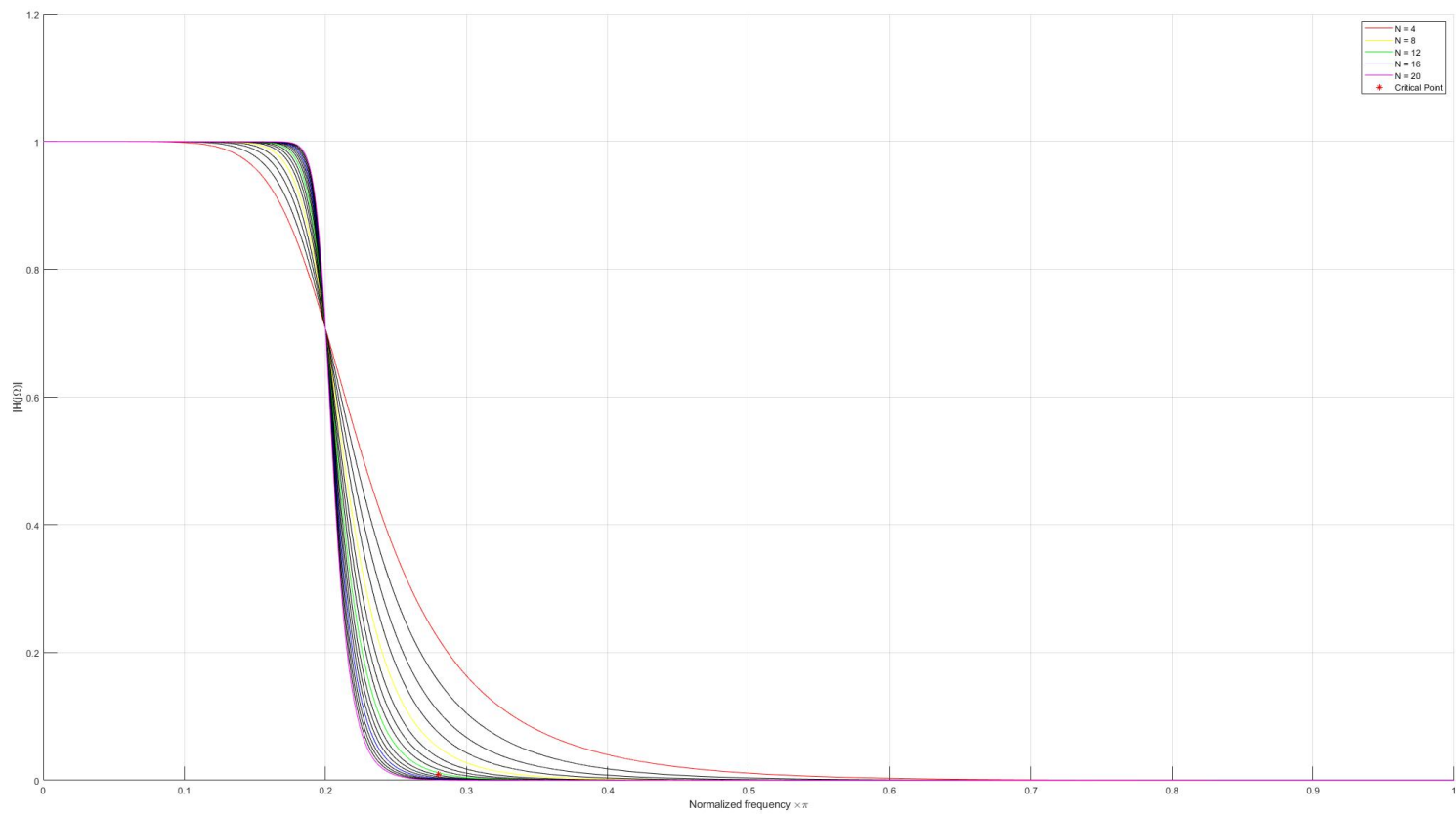
2. A Butterworth low-pass filter has the form

$$|H_c(j\Omega)|^2 = \frac{1}{1 + (j\Omega/j\Omega_c)^{2N}}$$

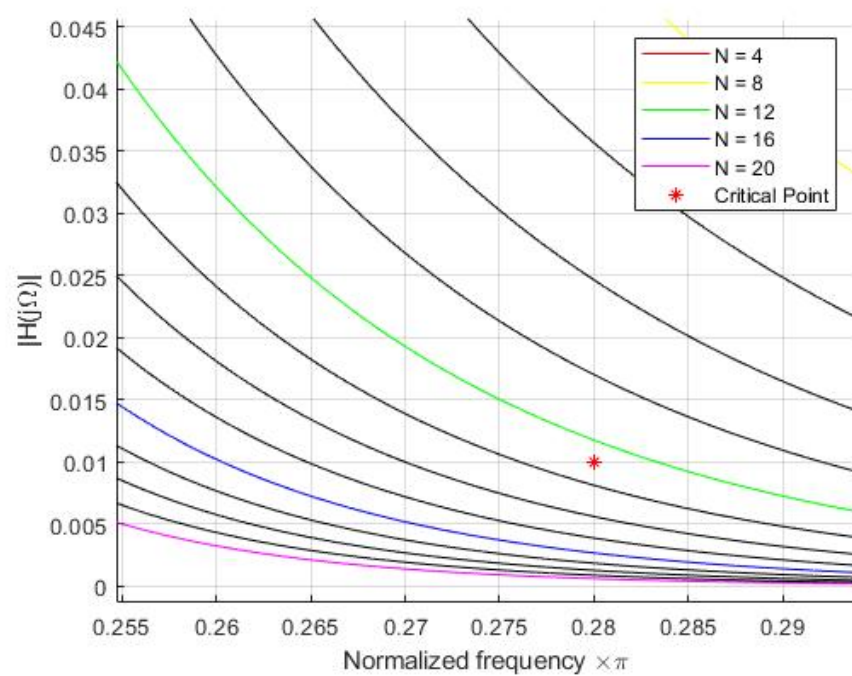
So we know that the magnitude response of a Butterworth low-pass filter monotonically falls off in the passband and stopband, it would always be equal to $\frac{1}{\sqrt{2}}$ or 0.7071 at the cut-off frequency because of the nature of the above form. The magnitude response of a 6th-order filter with $\Omega_c = 0.2\pi$ is shown on the right side. It drops to 0.707141 at about 0.2π as inference.



3. Initially, when $N = 4$, the red curve is smooth at the edge. As the parameter N increases from 4 to 20, according to the order of color of a rainbow, as the picture shows below. The filter characteristics become sharper. That implies it is more like an ideal LPF as N approaches to infinity.



4. Zoom in the region around 0.28π . Obviously, it can achieve our goal when N is greater than 12, so the answer is 13.



5.

Define

lower bound of frequency : lb

upper bound of frequency : ub

passband magnitude response : pb

stopband magnitude response : sb

The bilinear transformation will map the continuous-time frequencies back to the corresponding discrete-time critical frequencies. That is

$$pb \leq |H_c(j\Omega)| \leq 1, \quad 0 \leq \Omega \leq \frac{2}{T_d} \tan(lb/2)$$

$$|H_c(j\Omega)| \leq sb, \quad \frac{2}{T_d} \tan(ub/2) \leq \Omega \leq \infty$$

Since a Butterworth filter has a monotonic magnitude response, we can equivalently require that

$$|H_c(j2\tan(lb))/T_d| \geq pb$$

$$|H_c(j2\tan(ub))/T_d| \leq sb$$

Considering the Butterworth filter form mentioned in question 2, we obtain

```
function N = FunctionForQ5(lb,ub,pb,sb)
% passband ≤ |H(e^jw)| ≤ 1,      0 ≤ |w| ≤ lower bound
% |H(e^jw)| < stopband, upbound ≤ |w| ≤ π
% lb : lower bound
% ub : upper bound
% pb : passband
% sb : stopband

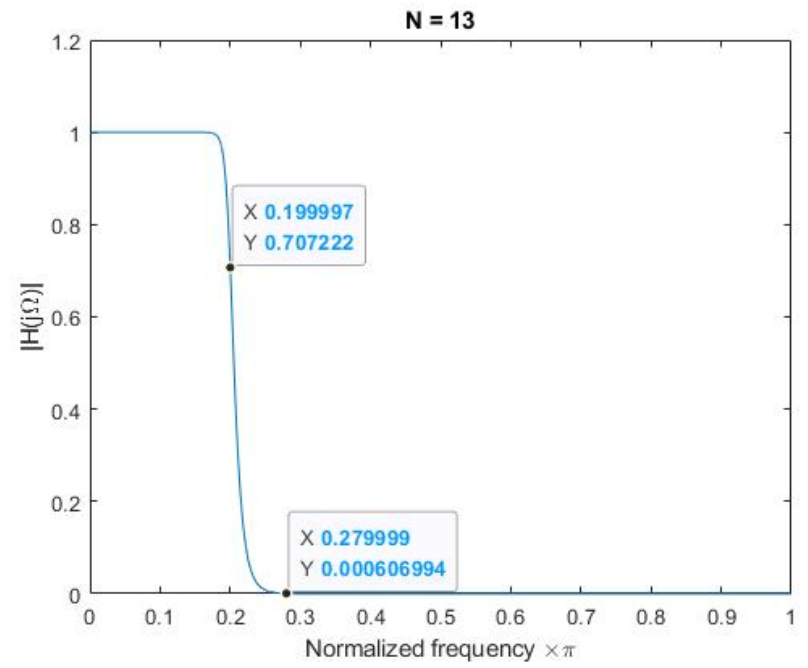
N = ceil(log(((1/sb)^2-1)/((1/pb)^2-1))/(2*log(tan(ub/2)/tan(lb/2))));
end
```

$$1 + \left(\frac{2\tan(lb/2)}{T_d\Omega_c}\right)^{2N} = \left(\frac{1}{pb}\right)^2$$

$$1 + \left(\frac{2\tan(ub/2)}{T_d\Omega_c}\right)^{2N} = \left(\frac{1}{sb}\right)^2$$

Solve for N. Note that N is required to be an integer.

$$N = \text{ceil}\left(\frac{\log\left[\left(\left(\frac{1}{pb}\right)^2 - 1\right) / \left(\left(\frac{1}{sb}\right)^2 - 1\right)\right]}{2\log\left[\tan(lb/2) / \tan(ub/2)\right]}\right)$$



Assume that passband is 0.2π , stopband is 0.28π , and the associated magnitude response is 0.7071 and 0.001, so N derived from FunForQ5 is 13. The result N is the same as that observed directly in question 3, and its corresponding magnitude response is shown above which suffices our design.

6.

The form of $H(z)$ can be represented as

$$H(z) = \frac{Y(z)}{X(z)} = C \prod_{k=1}^N \frac{(1 + q_k z)}{(1 - p_k z)},$$

where C is a constant.

```
function y = FunForQ6(NUM,DEN,x)
    ytermLen = length(DEN(2:end));
    xtermLen = length(NUM);
    x = reshape(x,[1,length(x)]);
    y = zeros(size(x));
    for ii = xtermLen:length(x)
        y(ii) = (sum(NUM.*fliplr(x(ii-xtermLen+1:ii)))-sum(DEN(2:end).*fliplr(y(ii-ytermLen:ii-1))))
    end
end
```

Reformulate z terms into

$$z^N Y(z) = (c'_N z^N + c'_{N-1} z^{N-1} + \dots + c')X(z) + (c_{N-1} z^{N-1} + c_{N-2} z^{N-2} + \dots + c)Y(z)$$

$$y[n] = c'_N x[n] + c'_{N-1} x[n-1] + \dots + c' x[n-N+1] + c_{N-1} y[n-1] + c_{N-2} y[n-2] + \dots + c y[n-N]$$

7.

A file named *boxing.wav* in HW4 is reused to test here. Before processing, it has been a clip with appropriate length. Now we only need to send this clip to FunForQ6. The parameters of Butterworth filter are referred to question 5 and its magnitude response is shown on the left side above. To see the differences between the original signal and signal applied LPF, their spectra are drawn in the same figure on the right side. Let's convert normalized frequency to Hz, that the cut-off frequency 0.2π is equal to $0.2 \times f_s / 2$, where f_s is the sampling frequency, e.g. 44100Hz. So it should have an obvious drop at about 4410 Hz theoretically. As the right figure shows, frequencies over about 5 kHz are suppressed fiercely. We further zoom in the region around 4410 Hz to justify our inference, that is shown below. The steep slope does credit to the high-order N such that frequencies over 4410 Hz can be reduced quickly.

